Supplier Decision Report

Milk procurement analysis · 10-week horizon · Generated 01 Nov 2026

2,000 litres/week · Stockout when delay > 20 min · Penalty ¹15,000/week

COMPARATIVE METRICS

METRIC	SUPPLIER A	SUPPLIER B
Mean delivery delay	14.30 min	**14.00 min**
Median delivery delay	14.50 min	**7.50 min**
Standard deviation (population)	**1.55 min**	11.30 min
Coefficient of variation	**10.86%**	80.69%
Karl Pearson's coefficient of skewness	**-0.386**	1.726
Stockout weeks (delay > 20 min)	**0**	3
Purchase cost per week	¹96,000	**¹92,000**
Total purchase cost (10 weeks)	¹9,60,000	**¹9,20,000**
Total stockout loss	**¹0**	¹45,000
Net cost (10 weeks)	**¹9,60,000**	¹9,65,000

RECOMMENDATION

Recommended supplier: Supplier A

Net cost ¹9,60,000 for 10 weeks · saves ¹5,000 versus Supplier B

Lower net cost: Supplier A lands at ¹9,60,000 for 10 weeks — ¹5,000 less than Suppl stockout losses are added to the purchase bill.

Lower coefficient of variation: CV of 10.86% versus 80.69% — delivery timing is 7.4× more predictable.

Lower delivery variability: Population standard deviation of 1.55 min against 11.30 min, so weekly scheduling is far easier to plan around.

Fewer stockout weeks: 0 weeks above the 20-minute limit versus 3 — avoiding ¹45,000

Supplier B's ¹46/litre rate is ¹2/litre cheaper, saving ¹40,000 on milk alone — b 3 stockout weeks add ¹45,000, leaving it ¹5,000 more expensive overall.

COST BREAKDOWN (10 WEEKS)

Purchase cost	¹9,60,000	¹9,20,000
Stockout loss	¹0	¹45,000
Net cost	**¹9,60,000**	**¹9,65,000**

Delay series · Supplier A: 12, 15, 14, 13, 16, 15, 14, 12, 17, 15 min · Supplier B: 5, 8, 30, 6, 7, 35, 9, 6, 28, 6 min
Mean, population standard deviation, CV and Karl Pearson skewness computed from the 10-week delay series.